LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Fuel contamination with water, foreign particles, other types of fuel, micro biological growth, contamination detection and control
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Fuel contamination with water, foreign particles, other types of fuel, micro biological growth, contamination detection and control
5. Specific Objectives	23.1.1.1 Trainee will be able to understand the construction and requirements of typical aircraft fuel system and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstrate fuel sample inspection using test kit; Safety: Wear gloves and eye protection; Supervision: Instructor monitors handling of samples
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary components of a typical aircraft fuel system?
(b) Why is fuel cleanliness critical for engine operation?
(c) List three common sources of fuel contamination.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Water contamination types and sources • Foreign particle contamination mechanisms • Fuel type incompatibility issues • Micro biological growth characteristics • Impact of contamination on fuel systems	L & Ds	PPT, IB	15 min
(b)	• Contamination detection methods and equipment • Water separation and filtration systems • Fuel sampling procedures and protocols • Prevention strategies for microbial growth • Maintenance schedules for fuel quality	L & Ds	PPT, PDF	20 min
(c)	• Emergency response to fuel contamination • Reporting procedures for fuel quality issues • Record keeping for fuel maintenance • Review of contamination control best practices • Summary of key learning points	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between free water and suspended water in fuel.
(b) Describe the procedure for identifying microbial growth in fuel samples.
(c) Outline the steps required to report a fuel contamination incident.

12. Home Assignment and Follow Up	Prepare a brief report on the specific fuel system components discussed in LP 02 that are susceptible to contamination and how they should be protected.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Hydraulic system, general design requirements, Typical layout of a basic hydraulic system and its brief description	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min